

 An  RPG Company	ROJA E	Email Id – roja.e@zensar.com Ph: 8197745717 Location: Bangalore LinkedIn: https://www.linkedin.com/in/roja-e-740718147	
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Summary	Senior Software Engineer with 5.10 years of experience in C/C++ development, specializing in system-level programming, multithreading, and network applications. Additionally, experienced in developing RESTful APIs using ASP.NET Core, with strong knowledge of C# concepts such as LINQ, delegates, and collections, and implementing secure solutions using JWT authentication and authorization.
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Education	Dayananda Sagar College of Engineering, B.E. in Electrical and Electronics Engineering (EEE) – [2015–2019]
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Core competencies/skills	Category	Primary skills	Secondary skills
	Domains	Navigation, GNSS	System software
	Tools	WSL/LINUX Visual Studio (2010/2019 /2022), Git Tortoise SVN,	Visual C++ 6.0 ,ReMa
	Databases and Backend	SQL, MYSQL	REST API Development, ASP.NET Core, Dependency Injection, Middleware
	Bug/Issue-Tracking System	Smart Tracker, Bugzilla	
	Programming skills	C++ (C++11/14/20/23), STL, C, CPP, Design Patterns, DSA	C#, .NET Core, ASP.NET Core Web API, Python (Basic),

Expertise	- Developed Windows desktop applications using C++, MFC, and VC++. - I worked on TCP/UDP-based network applications for real-time data communication. - Designed and implemented GUI-based tools for GNSS receiver analysis. - Worked on TCP/UDP-based network applications for real-time data comm.
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Experience Summary	Software Engineer PRDC [2019 -2021] - Skills: C/C++, MFC, SQL, MS-ACCESS, - Achievements: Implementation of data storage and fetch for the relays. Bug fixes on the modules. Performance improvement in a client server environment, old queries replacement with new queries.
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	• Software Engineer ALTEN GT [2021 to 2023] ○ Skills: C++, MFC, VC++, WIX, InstallShield, Socket Program ○ Achievements: Implemented TCP/UDP-based real-time communication systems. • Senior Software Engineer ALTEN GT [2023 to 2025] ○ Skills: C++, MFC, Dock Light ○ Achievements: Developed interactive GUI for analyzing data from Accord GNSS products. ○ Supported GPS, GLONASS, NavIC, BeiDou, GALILEO, and SBAS constellations.
Certifications	• completing the CompTIA Linux+: Installation & File System Navigation • Getting Started with GitHub Copilot • Python object-oriented programming with AI

PROFESSIONAL EXPERIENCE

Trade Processing and Settlement Engine (POC) | C++, Linux at Zensar Technologies (Mar 26 -Present)

Project Description:

Developed a Trade Processing & Settlement Engine using Modern C++ on Linux to simulate the post-trade lifecycle including trade validation, allocation, and settlement. The system demonstrates efficient processing of financial transaction actions and scalable backend design for handling trade workflows.

Key Activities Performed:

- Designed and implemented core modules using Modern C++ (C++11/14/17) with object-oriented programming and STL.
- Developed trade validation logic to ensure correctness of trade details such as quantity, price, and account information.
- Implemented trade allocation mechanisms to distribute trades across multiple accounts based on predefined rules.
- Built a settlement processing module to simulate trade settlement workflows, including updating account balances and positions.
- Designed data structures to efficiently manage trade records, account details, and transaction history.
- Applied multi-threading techniques to process multiple trades concurrently and improve system throughput.
- Implemented error handling and logging mechanisms to track trade failures and ensure data consistency.
- Designed an end-to-end trade lifecycle flow, from trade input to settlement completion, mimicking real-world financial systems.

Technologies / Skills / Tools Used:

VS Code, CMake , GiT .

Employee Management System – ASP.NET Core Web API (POC)

Project Description:

Developed a RESTful Web API using ASP.NET Core to manage employee data with secure authentication and clean architecture.

Key Activities Performed:

- Developed CRUD APIs using ASP.NET Core Web API
- Implemented JWT-based authentication and role-based authorization
- Used Dependency Injection for loose coupling and better testability
- Applied LINQ for efficient querying and data transformation
- Designed APIs using IActionResult and IActionResult<T>
- Implemented middleware pipeline for request handling
- Used collections for in-memory data management
- Tested APIs using Postman

Technologies:

C#, ASP.NET Core, Web API, JWT, LINQ

Project – MI Power / MiDBase at PRDC - [Aug 2019 to Oct 2021]

Role: Software Engineer

Project Description: MI Power Is a software comprehensive, Windows based Tool for power system analysis, planning and design. User friendly GUI, Centralized database, and handle complex system modeling. MiDBase is software for storing the user modeled data for the electrical network on which later the user is allowed to perform various network studies.

Key activities performed:

- Implementation of data storage and fetch for the relays. Bug fixes on the modules.
- Performance improvement in a client server environment, old queries replacement with new quires.
- Designing a Windows-based power system analysis and planning tool.

Technology/Skills/Tools used

- Tools: Bugzilla, Win Merge, Woises, Microsoft Access Database, SQL Server.
- Technologies: C++, MFC, SQL
- Skills: Functional testing, Setup Generation

Project –SMART WORKS and REMA at ALTEN Global Technologies Private Limited [Jan 2022 to May 2025]

Role: Sr. Software Engineer

Project Description: Smart Works is a client -Server based software which allows people in a work group to plan, manage and keep track of information flow. Requirement manager Is a Windows-based application to manage Requirement, design and test plan.

Key activities performed:

- Developed modules for project planning, task management, defect tracking, and risk management.
- Migration of REMA Client code Visual Studio 2008 to Visual Studio 2019.
- Enabled requirement for traceability, baseline, and audit support.
- Resolving Crashing issue. Generated server & client setup file.

Technology/Skills/Tools used

- Tools: Tortoise SVN, Smart Tracker
- Technologies: C++, MFC, VC++
- Skills: Functional testing, Setup Generation.

Project – GNSS RECEIVER GUI at ALTEN Global Technologies Private Limited [Nov 2021 to May 2025]

Role: Sr. Software Engineer

Project Description: DF-IG2SVisionUI GUI is a windows-based user interface for DF-IG2S dual frequency Receiver. It is a powerful tool for evaluating the performance characteristics of the receivers.

Key activities performed:

- Developed interactive GUI for analyzing data from Accord GNSS products.
- Supported GPS, GLONASS, NavIC, BeiDou, GALILEO, and SBAS constellations.
- Designing a Windows-based power system analysis and planning tool.

Technology/Skills/Tools used

- Tools: Tortoise SVN, Dock Light.
- Technologies: C++, MFC, VC++
- Skills: Functional testing, Setup Generation